

Lecture 7: Ethical Issues in Engineering Practice

HUM 4441: Engineering Ethics

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The Therac-25 Case

Between June 1985 and January 1987, at least six patients receiving radiation treatment from the Therac-25 were severely overdosed.

In one accident, the machine displayed Malfunction 54, reported an underdose, and paused. The manual did not explain the code. Believing the treatment was incomplete, the operator resumed it.

The patient had already received a massive overdose. A broken video monitor prevented the operator from seeing his reaction.

Decide before continuing

- What should the operator do?
- What information should the machine provide?
- Which failures belong to the engineers rather than the operator?

A software error became physical harm.

Fleddermann, Ch. 7, pp. 124, 131–133

Three Areas of Engineering Practice

Chapter 7 examines ethical problems that arise in three areas:

Environmental ethics

What responsibility does engineering have toward human health, communities, and the environment?

Computer ethics

How should engineers use computers and design systems that depend on software?

Ethics and research

What does honesty require while investigating and reporting results?

These are recurring parts of engineering work, not exceptional situations.

Fleddermann, Ch. 7, p. 125

Environmental Ethics and Sustainable Design

Sustainable design

- limits environmental harm during use
- considers manufacturing and disposal
- aims to maintain environmental quality and quality of life

Moral standing

- Anthropocentric view: nature matters through its value to humans.
- Broader view: animals, plants, or the biosphere may have value in themselves.

The chosen view changes what counts as environmental harm.

Fleddermann, Ch. 7, pp. 125–126

Two Approaches to Environmental Problems

	Cost-oblivious	Cost-benefit
Goal	Make the environment as clean as possible.	Balance pollution reduction against its costs.
Ethical connection	Rights and duty ethics	Utilitarianism
Main difficulty	“As clean as possible” is difficult to define, fund, and enforce.	Lives, species, and scenic value are difficult to price.

Justice problem: Neither approach is adequate if one group receives the benefits while another bears the environmental cost.

Cost-benefit analysis must ask who pays and who benefits.

Fleddermann, Ch. 7, pp. 126–127

The Albuquerque–Isleta Pueblo Water Case

- Isleta Pueblo lies downstream from Albuquerque and traditionally used Rio Grande water directly in religious ceremonies.
- Albuquerque discharged treated water that met applicable EPA standards.
- Isleta sought an arsenic standard roughly twice as strict as the EPA requirement.
- Albuquerque argued that compliance would cost approximately \$300 million and that the stricter standard was unnecessary.
- The EPA supported Isleta; the U.S. Supreme Court declined Albuquerque's appeal.

Should legal compliance settle the engineer's recommendation when a downstream community requires cleaner water?

Fleddermann, Ch. 7, pp. 137–138

The Engineer's Environmental Responsibility

Professional duties

- follow applicable laws and regulations
- protect public safety and the environment
- express concerns about unacceptable environmental effects

Professional responsibility includes knowing when another field's expertise is required.

Limits of competence

- Engineers may not be qualified to judge every environmental consequence.
- Seek help from biologists, physicians, public-health experts, and others.

Fleddermann, Ch. 7, p. 127

Computer Ethics: Two Categories

Category	Examples	Central issue
Computer used to commit an unethical act	theft, hacking, privacy invasion, copyright violation	The computer makes familiar wrongdoing easier, larger, or harder to trace.
Computer used improperly as an engineering tool	unsuitable design software, unchecked output, unsafe embedded control	The engineer remains responsible for the resulting design or system.

Computers create new ways for familiar ethical problems to occur.

Fleddermann, Ch. 7, pp. 127–130

Privacy, Hacking, and Copyright

Privacy

A person's ability to control access to and use of information about themselves. Centralized computer records make access easier for both authorized and unauthorized users.

Hacking

Unauthorized access remains unethical even when the hacker seeks challenge rather than harm. Damage can include lost records, money, work, or safety-critical information.

Copyright

Computers make unauthorized copying easy and difficult to detect; ease of copying does not remove ownership rights.

Fleddermann, Ch. 7, pp. 128–129

Computers as Design Tools

- CAD/CAM, circuit analysis, finite-element analysis, structural analysis, modeling, and testing software improve efficiency.
- Software can contain flaws or be unsuitable for a particular problem.
- Easy-to-use software can create an illusion of competence outside the engineer's field.

Fleddermann's requirements

- know the software's limits
- use current versions and patches
- verify computer-generated results

Software can never substitute for good engineering judgment.

Fleddermann, Ch. 7, pp. 129–130

Computers Integrated into Engineered Systems

- Embedded computers can improve efficiency and safety.
- Software control may remove humans from the control loop.
- A malfunction may occur without giving the operator useful warning.
- Poor design may prevent the operator from intervening.

Engineers should ensure that:

- software is adequately tested,
- humans can intervene when necessary, and
- safety does not rely solely on software when hardware redundancy is needed.

Fleddermann, Ch. 7, p. 130

Autonomous Computers

Autonomy is the ability of a computer to make decisions without human intervention.

- Autonomous control can be valuable in manufacturing and other fast processes.
- Automated stock selling contributed to the October 19, 1987 market crash: falling prices triggered selling, which drove prices lower and triggered more selling.
- Autonomous weapons raise greater concerns because instability or sensor failure can cause loss of life.

Where failure can become disastrous, meaningful human control remains necessary.

Fleddermann, Ch. 7, pp. 130–131

Computer Codes of Ethics

- Organizations have developed codes for the ethical use of computing resources.
- These codes serve the same purposes as other professional codes.
- They provide guidance for proper computer use.
- They do not replace sound moral reasoning and judgment.

A code can guide a decision, but the engineer must still understand the system and its consequences.

Fleddermann, Ch. 7, p. 131

Therac-25: What Went Wrong?

Software bug

Fast correction of treatment data could allow the beam to activate before the hardware reset.

Missing protection

Earlier machines used hardware interlocks; the Therac-25 relied heavily on software.

Weak communication

Error messages were unclear, and documentation did not explain them adequately.

Development process

Software modules and the complete software system were inadequately documented and tested.

The investigation identified both systems-engineering and software-engineering failures.

Fleddermann, Ch. 7, pp. 131–133

Therac-25: Assign Responsibility

The operator entered incorrect treatment information, corrected it quickly, and resumed after an unexplained error. The machine lacked a hardware interlock, reported an underdose, and provided no useful explanation of the malfunction.

Make a defensible judgment:

- ① Which operator mistakes should the design reasonably anticipate?
- ② Which single design change would have prevented the overdose?
- ③ What evidence would justify returning the machine to service?

Fleddermann, Ch. 7, pp. 131–134

Avanti Corp. vs. Cadence Design Systems

Cadence alleged that Avanti incorporated as many as 60,000 lines of Cadence code into competing electronic-design automation products. The indicted Avanti employees were former Cadence employees.

An employee may carry

general skills, experience, and professional judgment

An employee may not carry

source code, trade secrets, or proprietary information belonging to the former employer

Difficult boundary

knowledge learned through work that is not easily separated from protected intellectual property

Fleddermann, Ch. 7, pp. 134–135

Honesty in investigation

- avoid preconceived conclusions
- change the hypothesis when evidence requires it
- maintain objectivity

Honesty in reporting

- report results accurately
- do not overstate conclusions
- give proper credit
- clarify results later found to be incorrect

Virtue ethics identifies honesty as the central research virtue.

Fleddermann, Ch. 7, pp. 135–136

Irving Langmuir used the term for research in which investigators deceive themselves rather than intentionally commit fraud.

- effect caused by a barely detectable agent
- effect remains near the limit of detectability
- claims of great accuracy
- fantastic theories contrary to experience
- criticism answered with ad hoc excuses
- supporters rise, then gradually disappear

Subtle effects require strong controls, criticism, and independent confirmation.

Fleddermann, Ch. 7, pp. 136–137

The N-Ray Case

René Blondlot claimed to detect a new form of radiation through subtle changes in the brightness of a spark. Many laboratories could not reproduce the result.

During a visit, physicist R. W. Wood secretly removed a crucial prism from the apparatus. Blondlot's team continued to report the expected effect even though the prism was absent.

Discuss: Once Wood's result was known, what should Blondlot have done before making any further claim?

The case illustrates how expectation can shape observation.

Fleddermann, Ch. 7, pp. 138–139

Cold Fusion at Texas A&M

After the Pons–Fleischmann cold-fusion announcement, a Texas A&M group reported tritium production in electrochemical cells.

- The surprising results appeared at unusually convenient times.
- The results could not be reproduced reliably.
- Questions arose about contamination, possible spiking, weak controls, and laboratory supervision.
- Investigations found serious problems but did not establish fraud.

A researcher must investigate evidence against a desired result as seriously as evidence supporting it.

Fleddermann, Ch. 7, pp. 139–142

Ghostwriting of Research Articles

- Research articles must accurately identify who performed and shaped the work.
- Hidden sponsor-linked writers prevent readers from judging conflicts of interest.
- Giving authorship credit to someone who did not make a substantial contribution is dishonest.
- Omitting a substantial contributor is also dishonest.

Decide: A company-paid writer drafts the literature review and shapes the conclusion, while a professor is listed as the only author. What must change before submission?

Fleddermann, Ch. 7, pp. 142–143

Summary

- Environmental decisions require attention to sustainability, moral standing, cost, and the distribution of burdens.
- Engineers remain responsible for computer-assisted designs and for safe integration of software into systems.
- Autonomous systems require human control when failure can cause serious harm.
- Therac-25 shows why software testing, hardware redundancy, documentation, and usable warnings matter.
- Research requires honesty in investigation, reporting, and attribution.

Reference

Charles B. Fleddermann, *Engineering Ethics*, Chapter 7, “Ethical Issues in Engineering Practice.”